Read, Not Reasoned: Small Language Models Copy From Their Reasoning Chains — Evidence From Ground-Truth Interventions

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Abstract

Does a language model's answer depend on the reasoning chain it writes, and through which path? We answer with interventions whose ground truth is known by construction: inserted redundant hops (true necessity 0), programmatic input changes (true effect 0), and exact mediation in which the chain is held fixed while the input is changed. Across two model families (Qwen2.5-1.5B, SmolLM2-1.7B) and two multi-hop datasets we find three things. **(1) Chains are read, not reasoned over.** When the answer is a string written in the chain (MuSiQue), the chain's content moves p(gold) by 56–72 points; when the answer must be composed from the chain (CLUTRR), by 1–2 points, and getting the chain right does not raise accuracy. Appending a single hop that writes the answer into an otherwise identical CLUTRR chain raises the content effect to 63–75 points. **(2) Step-masking tests are blind to bridge reasoning.** Masking one intermediate hop changes the answer by ≤ 1.3 points, while masking it together with its descendants in the chain's reasoning graph changes it by 66–76; total-effect estimators identify necessary hops with AUC 0.98–1.00 versus 0.64–0.83 for direct-effect ones, and placebo and mask-token controls stay at chance. **(3) Reasoning relocates vulnerability.** Distractor paragraphs cost chain-of-thought 7–9 EM points, entirely through the regenerated chain (natural indirect effect -9 to -10; direct effect with the chain fixed $\approx +1$), whereas direct answering is hurt at the readout. Consequently, readout-level contrastive decoding, including the graph-aware rule we propose (HGCD), cannot improve accuracy or robustness at this scale; its apparent gains on CLUTRR are matched by content-free label calibration. All hypotheses were fixed before each GPU run; all code runs on a free Kaggle GPU.

1 Introduction

Chain-of-thought prompting [33] makes models write intermediate steps, but writing a step is not the same as *using* it. Causal tests show that models frequently ignore or post-rationalise their own chains [9, 15, 28, 29], answer multi-hop questions through single-hop shortcuts [2, 13, 35], and are easily derailed by irrelevant context [20]. Two families of remedies exist. *Training* remedies (counterfactual preference objectives [8, 15], PNS-filtered supervision [36], causality-aware post-training [6]) are effective but costly. *Decoding* remedies are training-free: contrastive decoding [11, 14], classifier-free guidance [19], and context-aware decoding (CAD) [21] with its adaptive and gated successors [25, 31].

CAD scores tokens by $(1{+}\alpha)\log \pi(\cdot \mid c, x) - \alpha \log \pi(\cdot \mid x)$. Read causally, the log-ratio is the effect of do($C{=}c$) versus do($C{=}\varnothing$): the total effect of the *entire* context. In multi-hop QA the context contains supporting paragraphs on the causal path to the answer and distractors that are not; CAD amplifies both. This offers a causal explanation for the empirical failures of CAD on uninformative or distracting context that motivated AdaCAD and NWCAD [25, 31].

A decoding probe. Instead of contrasting *with versus without context*, contrast the model's answer distribution *given its own chain* against the distribution under *interventions on the hops of that chain*, keeping the context fixed. Because the context is held fixed in both terms, distractor and parametric components cancel; because interventions follow the chain's own graph, the contrast measures the *path-specific* contribution of the reasoning steps.

Contributions.

1. **A ground-truth protocol for chain reliance**: binary, programmatic interventions with known effects (redundant hops with necessity 0; renaming, reordering, distractors and neutral insertions with effect 0), placebo and mask-token controls, and an *exact* mediation analysis in which the chain is set directly.
2. **Copy, not compute (F1)**: small models depend on their chain when the answer is a string written in it and not when it must be composed from it; one binary intervention (T_{copy}) switches the dependence on, replicated across two model families.
3. **A blind spot of step-masking tests (F2, proposition 1)**: masking one step measures a controlled direct effect that vanishes on bridge chains; graph-aware total-effect interventions fix it, and two independent implementations agree.
4. **Reasoning relocates vulnerability (F3)**: distractors harm chain-of-thought only through the regenerated chain, and direct answering only at the readout.
5. **Limits of contrastive decoding (F4)**: we formulate HGCD, which contrasts the model with graph-aware interventions on its own hops, relate it to PNS (proposition 4) and CAD (proposition 3), and show why such readout-level rules cannot help when reliance is absent or the vulnerability lies upstream.

2 Related Work

Faithfulness of reasoning chains. Lanham et al. [9] intervene on model-written chains (truncation, inserted mistakes, paraphrases); Turpin et al. [28] show biased features steer answers without being verbalised; FRODO [15] uses causal mediation [30] and trains small models with counterfactual preferences; FUR [29] measures faithfulness by unlearning steps; Thought Anchors [3] measures step importance by counterfactual resampling, and Thought Branches [12] argues that step effects must be measured by resampling the continuation (a total effect), noting that removing a sentence while preventing its content from reappearing changes conclusions. FRODO already treats the rationale as a mediator and finds weak indirect effects for models under 20B, and Somov et al. [23] find that models often fail to update after edits to intermediate structures. Our contribution relative to these is narrower and specific: (i) necessity labels and null effects that are *known by construction*, so step-importance estimators can be *validated* rather than compared with each other; (ii) a deterministic, rollout-free total-effect intervention for hop-structured chains, validated against resampling; (iii) the bridge-versus-compositional distinction, which predicts exactly where direct-effect estimators fail; and (iv) an exact mediation of distractor effects (chain vs. readout).

Necessity and sufficiency. Yu et al. [36] define step-level probabilities of necessity and sufficiency for chain-of-thought, prove identifiability under exogeneity and monotonicity (building on Pearl [17], Tian and Pearl [26]), and use PNS to prune chains for in-context learning and fine-tuning. We show the same quantity arises as the limit of a decoding rule, and we separate direct from total effects [16].

Table 1: Positioning against closely related work. "GT" = the true value of the measured effect is known by construction.

Work	What is intervened on	Effect measured	GT?	Training	Controls
Lanham et al. [9]	model's CoT (truncate, add mistake, paraphrase)	answer change	no	none	paraphrase, filler tokens
Stolfo et al. [24] (ACL)	input factors of math problems	total / direct causal effect	partly	none	–
Paul et al. [15] (EMNLP-F)	rationale	mediation (indirect effect)	no	preference training	–
Yu et al. [36] (NeurIPS)	CoT step, replaced + rolled out	PNS	no (LLM judge)	SFT / ICL	–
Macar et al. [12] (ICLR)	sentence, resampled continuation	total effect, resilience	no	none	on- vs. off-policy
Somov et al. [23]	intermediate structure (rubric)	decision update rate	yes (deterministic map)	none	–
Tutek et al. [29] (EMNLP)	step information in weights	prediction change	no	unlearning	–
Shi et al. [21], Wang et al. [31]	whole context	(implicit) total context effect	no	none (decoding)	–
Ours	hop + DRG descendants; binary input interventions	direct vs. total effect; NDE/NIE	**yes**	none	placebo, mask-marker, redundant hop, content-free

Contrastive and guided decoding. Contrastive decoding [11, 14], DoLa [4], CFG [19], CAD and its adaptive/gated variants [21, 25, 31] contrast distributions under different models, layers or contexts. Visual contrastive decoding [10] uses a *distorted* rather than empty input; our L1 variant is the textual analogue with a type-matched counterfactual. Pced [5] treats each retrieved document as an expert contrasted with a no-context baseline; RED [34] multiplies image- and rationale-conditional distributions in vision-language models. None of these intervene on the hops of the model's own reasoning graph, distinguish direct from total effects, or relate the contrast to PNS.

Causal analyses of reasoning. Stolfo et al. [24] quantify robustness of mathematical reasoning through interventions on a causal graph of the input; Zečević et al. [37] argue models "talk" causality without being causal; text-intervention effect estimation [7] uses LLM rewrites as treatments. Our interventions are programmatic, so their ground-truth effects are known exactly.

3 Setup: a Structural Causal Model of Multi-Hop Reasoning

Variables. A question Q, a context $P = (P_{\text{sup}}, P_{\text{dist}})$ of supporting and distractor passages, a chain of hops $H_1, \ldots, H_k$, and an answer A. Each hop is rendered as `Hop` i`:` $\langle$`body`$\rangle$ `=>` $\langle$`result`$\rangle$. The model acts as a *reader* mechanism $\pi(a \mid Q, P, h_{1:k})$ over a finite candidate set $\mathcal{A}$.

Two task graphs. In *bridge* chains (MuSiQue), hop j resolves an entity that the next hop consumes and the answer is the result of the last hop: $Q \to H_1 \to \cdots \to H_k \to A$, with $P_{\text{sup},j} \to H_j$. In *compositional* chains (CLUTRR), every hop contributes a relation that is composed into the

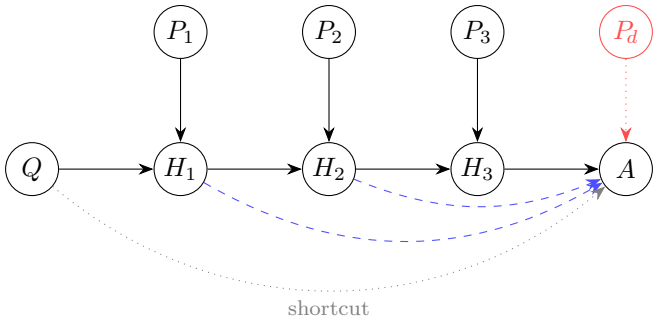


Figure 1: Task SCM. Solid: bridge chain. Dashed blue: extra edges of compositional chains. Red dotted: spurious distractor path that the *reader* may use but the task graph forbids. Gray: parametric shortcut. CAD's contrast includes the red path; HGCD holds P fixed and contrasts interventions on H_j only.

answer: $H_j \to A$ for all j and $H_j \to H_{j+1}$ through the shared entity. Distractors have no edge to A in either task graph (fig. 1).

Directed reasoning graph (DRG). Given a generated chain $y = (h_1, \ldots, h_k)$ we add an edge $i \to j$ $(i < j)$ if the result of h_i is mentioned in the body of h_j. Terminal hops are those whose result is the query target (CLUTRR) or the last hop (MuSiQue); the *relevant set* $\mathcal{R}(y)$ contains the terminal hops and their DRG ancestors. Hops outside $\mathcal{R}(y)$ have no directed path to the answer.

Interventions on hops.

Definition 1 (Naive and graph-aware interventions)**.** y_j^{nai} *replaces hop* j *by a mask token and leaves all other hops unchanged.* y_j^{tot} *masks hop* j *and, in every DRG descendant* d *of* j*, masks every mention of the results of* j *and of* d*. The latter is a deterministic, rollout-free approximation of* $\text{do}(H_j{=}\perp)$*: descendant values are functions of* h_j*, hence unknown under the intervention. A* placebo *rewrites hop* j *without changing its entities or relation.*

For an answer a we write $\Delta_j^{\text{nai}}(a) = \pi(a \mid y) - \pi(a \mid y_j^{\text{nai}})$ and $\Delta_j^{\text{tot}}(a) = \pi(a \mid y) - \pi(a \mid y_j^{\text{tot}})$, suppressing Q, P.

What is (and is not) causal here. We do not assume the language model *is* an SCM of the task. The reader π is a deterministic function of its input, so editing hops in the prompt is an exact intervention on the reader's input, and Δ_j is an exact (not estimated) effect *on the reader*. The task SCM only supplies ground truth: which hops are necessary, and which input changes should have no effect on the answer.

Mask-token artifacts. Replacing text by `[MASKED]` is itself out of distribution. Our estimator benchmark controls for this: the inserted redundant hop is masked in exactly the same way, so any effect caused by the mask token alone appears as a non-zero Δ on redundant hops and lowers the AUC. In addition, a *mask-marker* control inserts `[MASKED]` into hop j while keeping all of its content; its effect Δ_j^{marker} measures the mask-token artifact directly and should be ≈ 0.

Algorithm 1 HGCD (L2) for one question

$y \leftarrow$ greedy chain from the model (optionally with L1 contrast on result tokens)
Parse hops; build DRG; compute relevant set $\mathcal{R}(y)$
$p \leftarrow \pi(\cdot \mid Q, P, y)$ over $\mathcal{A}$ ▷ one forward + KV-cached candidates
for $j \in \mathcal{R}(y)$ **do**
    $p_j \leftarrow \pi(\cdot \mid Q, P, y_j^{\text{tot}})$
end for
return $\arg\max_{a \in \mathcal{V}_\beta} (1+\alpha) p(a) - \alpha \operatorname{mean}_j p_j(a)$

4 Method: Hop-Graph Contrastive Decoding

Answer-level rule (L2). With uniform weights $w_j = 1/|\mathcal{R}(y)|$ on relevant hops and a plausibility set $\mathcal{V}_\beta = \{a : \pi(a \mid y) \geq \beta \max_{a'} \pi(a' \mid y)\}$ [11],

$$\hat{a} = \arg\max_{a \in \mathcal{V}_\beta} s_\alpha(a),$$
$$s_\alpha(a) = (1+\alpha)\,\pi(a \mid y) - \alpha \sum_{j \in \mathcal{R}(y)} w_j\, \pi(a \mid y_j^{\text{tot}}) = \pi(a \mid y) + \alpha \sum_{j \in \mathcal{R}(y)} w_j\, \Delta_j^{\text{tot}}(a). \tag{1}$$

The score is the answer probability plus α times the average total effect of the answer-relevant hops on that answer. Redundant hops ($j \notin \mathcal{R}(y)$) are never contrasted.

Evidence-level rule (L1). While emitting the *result* tokens of a hop, we contrast the real context P with a type-matched counterfactual $\tilde{P}$ of identical length and entity types in which the relational content differs (on CLUTRR: names permuted within each gender group, so every sentence and entity type is kept but who-is-related-to-whom changes; we evaluate L1 on CLUTRR only, where $\tilde{P}$ is exactly type-matched): $t^\star = \arg\max_{t \in \mathcal{V}_\beta} (1+\alpha)\pi(t \mid P, \cdot) - \alpha\pi(t \mid \tilde{P}, \cdot)$. Unlike an empty context, $\tilde{P}$ is in-distribution and controls for length and type cues.

Remark 1 (What each level can fix)**.** *L2 makes the answer depend* more *on the chain; it therefore repairs* readout *errors (chain correct, answer wrong because a shortcut or distractor dominates the readout) but cannot repair a wrong chain—it may even commit harder to one. Chain errors are the target of L1, which acts while the hops are produced. We test this split directly (hypothesis 4).*

Cost. L2 needs $1 + |\mathcal{R}(y)|$ prefix forwards and a KV-cached pass over the candidate tokens; no extra generation. L1 doubles the forward cost of result tokens only.

Choice of α. α is selected on the CLUTRR validation split and frozen (ties broken toward the smaller α, so HGCD reduces to CoT absent evidence); an adaptive variant sets $\alpha \cdot \mathrm{JSD}(p, \bar{p}_j)$ as in AdaCAD [31].

5 Theory

All proofs are short and given in appendix A; the propositions make predictions that our protocol tests directly.

Assumption 1 (Markov reader on bridge chains)**.** *For bridge chains, $\pi(a \mid h_{1:k})$ depends on $h_{1:j}$ only through $h_{j+1:k}$ whenever h_{j+1} restates the result of h_j ($j < k$).*

Proposition 1 (Naive masking is a direct effect)**.** $\Delta_j^{\text{nai}}(a)$ *equals the controlled direct effect of* H_j *on* A *with* $H_{j+1:k}$ *held at their observed values. Under assumption 1,* $\Delta_j^{\text{nai}}(a) = 0$ *for every* $j < k$ *on bridge chains, whereas* $\Delta_j^{\text{tot}}(a)$ *is not constrained to zero. On compositional chains, where* A *reads each hop's relation, the direct effect is generally non-zero.*

Prediction. On CLUTRR both estimators separate necessary from redundant hops; on MuSiQue, $\Delta^{\text{nai}} \approx 0$ on non-final gold hops while Δ^{tot} remains large, so the gap between the two appears in the position breakdown of table 4.

Assumption 2 (Mixture reader)**.** $\pi(a \mid Q, P, y) = \lambda_c \pi_c(a \mid y, P_{\text{sup}}) + \lambda_s \pi_s(a \mid Q) + \lambda_d \pi_d(a \mid P_{\text{dist}})$*, with* λ*'s non-negative, summing to one, and invariant to interventions on hops.*

Proposition 2 (Exact cancellation in probability space)**.** *Under assumption 2,*

$$s_\alpha(a) = \pi(a \mid y) + \alpha\lambda_c \sum_j w_j [\pi_c(a \mid y) - \pi_c(a \mid y_j^{\text{tot}})],$$

i.e. the shortcut π_s *and distractor* π_d *components cancel in the contrast for every* α*. The log-space analogue does not cancel them.*

Proposition 3 (CAD amplifies distractor paths)**.** *Under assumption 2 applied to direct answering, with no-context distribution* $\pi_s(a \mid Q)$*, CAD's score is* $(1+\alpha)\log[\lambda_c\pi_c + \lambda_s\pi_s + \lambda_d\pi_d](a) - \alpha \log \pi_s(a \mid Q)$*. For any* $\alpha > 0$ *and any gold answer* g*, a distractor answer* a_d *with* $\pi_d(a_d) = 1$ *is preferred by CAD whenever* $\pi_s(a_d \mid Q)$ *is small enough, even if* $\pi(g \mid Q, P) > \pi(a_d \mid Q, P)$*. In contrast, HGCD's contrast term does not depend on* π_d *at all (proposition 2).*

Prediction. Under distractor injection, CAD's accuracy degrades more than HGCD's, i.e. $|\text{ATE}_{\text{distract}}|$ is larger for CAD.

Proposition 4 (PNS limit)**.** *Interpret* $\pi(a \mid y_j^{\text{tot}})$ *as* $P(A{=}a \mid \text{do}(H_j{=}\perp))$ *and assume exogeneity and monotonicity as in Yu et al. [36]. Then* $\Delta_j^{\text{tot}}(a) = \text{PNS}_j(a)$ *and* $\lim_{\alpha\to\infty} s_\alpha(a)/\alpha = \sum_{j\in\mathcal{R}(y)} w_j \,\text{PNS}_j(a)$*. Hence, within* $\mathcal{V}_\beta$*, HGCD interpolates between the most probable answer (*$\alpha{=}0$*) and the answer for which the relevant hops are, on average, most necessary and sufficient.*

6 Experimental Protocol

Data. CLUTRR [22] (`gen_train234_test2to10`), restricted to clean simple-path stories (the filter is reported), gives exact gold paths; ID $k \in \{2, 3, 4\}$, OOD $k \geq 5$; answers are scored over the 20 kinship relations. **MuSiQue-Ans** [27] gives real text with gold decompositions; the context holds the supporting paragraphs plus 4 distractors. MuSiQue was built to resist single-hop shortcuts [13], which is why we prefer it to HotpotQA. Candidates for MuSiQue are built without the gold answer: the model's own answer and hop results, then paragraph titles, capitalised spans and dates/numbers from the context, in order of appearance, capped at 64 (recall of the gold answer is 56% on 600 dev questions before adding the model's strings, versus 10.5% for titles alone). All methods share the same candidate set and we report its recall.

Binary interventions with known effects.

Intervention	Definition (programmatic, seeded)	True effect on A
`rename`	all names → fresh fictional names (gender-preserving)	0
`order`	shuffle story sentences / paragraphs	0
`distract` / `dist8`	insert a disconnected family / 4 extra distractor paragraphs	0
`placebo_input`	insert two neutral sentences without entities	0
redundant hop	insert a true but off-path fact as a hop of a gold chain	necessity 0

For each method we report the paired ATE $\mathbb{E}[Y(1) - Y(0)]$ on accuracy with bootstrap 95% CIs; a robust method has ATE ≈ 0. `placebo_input` controls for "any change hurts".

Estimator validation. Gold chains with one inserted redundant hop give hop-level labels (necessary = 1, redundant = 0). We report AUC of Δ^{tot}, Δ^{nai}, the placebo and mask-marker Δ, rollout PNS [36] and the DRG path indicator, overall and on chains the model answers correctly (PS = 1), and mean effects by hop position.

Methods. Direct answering; CoT; self-consistency [32] with 5 sampled chains (strictly more compute than HGCD, which generates nothing extra); CAD [21]; AdaCAD [31]; CAD-chain (contrast with vs. without the chain, a text analogue of rationale-conditioned decoding); HGCD and ablations: naive masking, no graph (all hops), log space, adaptive α, placebo contrast (the key control); L1 chains decoded with CoT or HGCD.

Hypotheses. The following hypotheses and decision criteria were fixed in the repository before any GPU experiment was run (verifiable in the commit history).

Hypothesis 1 (Estimator)**.** *AUC*$(\Delta^{\text{tot}}) \geq 0.75$ *on both datasets;* Δ^{nai} *near zero on MuSiQue non-final gold hops; placebo and mask-marker* $\Delta \approx 0$ *on all hops.*

Hypothesis 2 (Accuracy)**.** *HGCD > CoT and > CAD-chain on OOD CLUTRR (*$k \geq 5$*) and on MuSiQue, with paired bootstrap CIs excluding zero.*

Hypothesis 3 (Robustness)**.** $|\text{ATE}_{\text{distract}}|$ *of HGCD < that of CAD (proposition 3); ATE of* `placebo_input` ≈ 0 *for all methods.*

Hypothesis 4 (Readout)**.** *The gain of HGCD over CoT is concentrated on questions whose greedy chain reaches every gold-path entity, and is* ≈ 0 *(or negative) on the others (remark 1).*

Hypothesis 5 (Ablations)**.** *HGCD* $\geq$ *naive-mask and no-graph variants; probability space* $\geq$ *log space; and, crucially, HGCD > placebo contrast. Contrasting against* any *perturbation can move decisions (our CPU pilot shows this), so the placebo contrast, not CoT, is the control that isolates the causal content of the intervention.*

Models and compute. Qwen2.5-1.5B-Instruct [18] (main) and SmolLM2-1.7B-Instruct [1] (second family) on two Kaggle T4 GPUs in fp16; Qwen2.5-7B-Instruct on one A100 for the scale point. Greedy decoding; $\beta = 0.1$; $\alpha \in \{0, 0.25, 0.5, 1, 2, 4\}$ chosen per method on the CLUTRR validation split and transferred unchanged to MuSiQue (no MuSiQue data is used for tuning).

Statistics. Bootstrap CIs (2,000 resamples), paired bootstrap differences and exact McNemar tests on the same questions. The three *primary* comparisons are HGCD vs. CoT, vs. CAD-chain and vs. the placebo contrast, on OOD CLUTRR and on MuSiQue; all other contrasts are exploratory and are reported without multiplicity claims. With n questions per split, the half-width of a 95% CI on accuracy is about $1/\sqrt{n}$; paired tests are considerably tighter.

7 Results

All tables are produced by the released notebook and `tools/summarize_runs.py`*; nothing is typed by hand.* Setup: two models (Qwen2.5-1.5B-Instruct, SmolLM2-1.7B-Instruct), fp16, one Kaggle T4 each; 238 CLUTRR test questions (118 ID, 120 OOD), 120 MuSiQue questions; estimator benchmark on 90 CLUTRR and 80 MuSiQue gold chains (one inserted redundant hop each).

Table 2: Main quantities, replicated across two model families (effects and accuracies in points; AUC for separating necessary from redundant hops).

Quantity	qwen-1.5b	smollm2-1.7b
Content effect of chain, CLUTRR	0.8	2.0
Content effect, CLUTRR + answer hop (T_{copy}=1)	75.1	62.8
Content effect, MuSiQue	56.4	71.9
CLUTRR+copy non-final hops: Δ^{tot} / Δ^{nai}	76.3 / 1.3	68.5 / 1.1
MuSiQue non-final hops: Δ^{tot} / Δ^{nai}	66.3 / 0.7	72.8 / -3.6
AUC MuSiQue: total / direct	0.975 / 0.828	0.988 / 0.682
AUC CLUTRR+copy: total / direct	1.000 / 0.658	0.991 / 0.643
MuSiQue `dist8`, CoT: TE	-9.2	-7.5
NDE (chain fixed) / NIE (via chain)	0.8 / -10.0	1.7 / -9.2
MuSiQue `dist8`, direct answer: NDE	-4.2	-6.7
CLUTRR accuracy: calibration − CoT	5.9	3.4

F1: chains are read, not reasoned over (H11 supported). The chain's content effect on p(gold) (real chain minus a chain of the same shape with every hop masked) is 56–72 points on MuSiQue, where the answer is the result written in the last hop, but only 0.8–2.0 points on CLUTRR, where the answer is a relation that must be composed from the hops and never appears in them (table 2). The binary intervention T_{copy}, which appends one hop stating the answer and leaves everything else identical, raises the CLUTRR content effect to 75.1 (Qwen) and 62.8 (SmolLM2) points. Two further facts point the same way: on the model's own chains, reaching every gold-path entity does not raise CLUTRR accuracy (Qwen: 17.1% with gold-path chains vs. 18.3% otherwise; SmolLM2: 15.2% vs. 22.4%; table 8), and the chain's *form* has no effect (0.2 and −2.1 points; table 5). At this scale a chain functions as a place to look up an answer, not as a computation the model carries out.

F2: direct-effect estimators miss bridge steps (H1, H-validity supported). Whenever the answer is copied (MuSiQue, CLUTRR with T_{copy}), masking one non-final hop changes p(gold) by −3.6 to 1.3 points, while masking it with its DRG descendants changes it by 66–76 points, close to the effect of the final hop. Two independent total-effect implementations (masking, counterfactual replacement with propagation) agree (AUC 0.975–1.000), and both direct variants fail (0.64–0.83). Placebo rewrites and the mask-marker control stay at chance on every dataset and model (table 4), so the effects are not artifacts of the mask token or of rewording.

F3: reasoning relocates vulnerability (H6 supported; proposition 3 not supported). On MuSiQue, distractors lower CoT's exact match by 9.2 (Qwen) and 7.5 (SmolLM2) points. Holding the original chain fixed removes the entire effect (NDE +0.8 and +1.7); it flows through the regenerated chain (NIE −10.0 and −9.2; table 6). Every chain-based reader, including HGCD, CAD-chain, calibration and the direct×CoT product, shows the same pattern. Direct answering

has no chain to corrupt but is hurt at the readout (NDE -4.2 and -6.7). With the chain fixed, CAD's readout is no more fragile than direct answering, so the mechanism of proposition 3 does not appear at this scale.

F4: decoding cannot amplify dependence that does not exist (H2 not supported; H8 supported). HGCD matches CoT on both datasets and models ($|\Delta| \leq 1.7$ points, all $p \geq 0.6$). The only decoding rule that beats CoT on CLUTRR is content-free label calibration (Qwen +5.9 points, McNemar $p = 0.009$ against HGCD; SmolLM2 +3.4, n.s.), which uses no information about the chain at all and is itself fragile (ATE of the neutral-sentence placebo -5.0 $[-10.1, -0.4]$ for SmolLM2). The earlier gains of the naive-mask and log-space variants at $\alpha = 4$ are of this kind. CLUTRR errors at this scale are dominated by the label prior, which F1 explains: the model is not using the chain to compute the relation.

Table 3: Hypothesis scorecard after two GPU rounds.

Hypothesis	Evidence (both models)	Verdict
H1 estimator validity (copyable answers)	AUC 0.975–1.000 total vs. 0.64–0.83 direct	✓
H11 copy, not compute	content 1–2 $\rightarrow$ 63–75 points with T_{copy}	✓
H6 mediation	dist8: NIE -9 to -10, NDE $\approx +1$ for every chain-based reader	✓
H8 calibration explains decoding gains	calibration $\geq$ naive/log variants	✓
H2 HGCD improves accuracy	= CoT everywhere	×
H3 / proposition 3 CAD more fragile at readout	CAD NDE $\approx$ direct NDE	×
H-form chain form explains CoT gains	form effect 0.2 / -2.1 points	×
H4 readout	gold-path chains do not raise accuracy (supports F1, not HGCD)	–
H-scale, H-niche	require the scale study (`MODE="reliance"`)	pending

7.1 Round-3 hypotheses (fixed before the next run)

Hypothesis 6 (Does computation emerge with scale?)**.** *The CLUTRR content effect (without T_{copy}) stays below* 10 *points up to 3B and rises above it at 7B; the gap between total and direct estimators on bridge chains persists at every scale.*

Hypothesis 7 (Copying on the model's own chains)**.** *When the gold relation word appears in the model's own CLUTRR chain, CoT accuracy is higher and the chosen relation is a word from the chain more often than when it does not.*

Hypothesis 8 (Copying overrides the chain's own logic)**.** *If the appended hop states a* wrong *relation that contradicts the relational hops before it, the model follows it in most chains (argmax equals the wrong relation in $> 50\%$), i.e. it reads the stated answer instead of checking it against the hops it reads past. A model that computes over the chain would resist.*

Hypothesis 9 (Cheap total effect)**.** *Rollout PNS and graph-aware masking rank hops consistently (AUC difference ≤ 0.05), at about $1/K$ of the cost.*

Table 4: Estimator validation, Qwen2.5-1.5B (SmolLM2 in appendix C).

Estimator	clutrr		clutrr_copy		musique	
	all	PS=1	all	PS=1	all	PS=1
Mask hop only (direct)	0.532	0.583	0.658	0.664	0.828	0.838
Mask hop + DRG descendants (total)	0.523	0.539	1.000	1.000	0.975	0.976
Replace result, keep downstream (direct)	0.582	0.650	0.672	0.670	0.764	0.767
Replace result, propagate (total)	0.576	0.635	1.000	1.000	0.978	0.978
DRG path indicator (structure only)	1.000	1.000	0.500	0.500	0.994	0.994
Placebo rewrite (control)	0.468	0.469	0.498	0.509	0.456	0.450
Mask token, content kept (control)	0.542	0.558	0.501	0.515	0.505	0.517

Dataset	Hop type	mask-tot	mask-dir	cf-tot	cf-dir	placebo	marker
clutrr	gold-final	0.004	0.004	0.043	0.043	0.006	0.015
clutrr	gold-nonfinal	-0.017	0.000	0.007	0.010	-0.002	-0.010
clutrr	redundant	-0.005	-0.005	-0.009	-0.009	0.009	-0.006
clutrr_copy	gold-final	0.721	0.721	0.782	0.782	-0.001	0.005
clutrr_copy	gold-nonfinal	0.763	0.013	0.785	0.009	0.002	0.008
clutrr_copy	redundant	0.005	0.005	-0.003	-0.003	-0.000	0.006
musique	gold-final	0.690	0.690	0.907	0.907	-0.002	0.002
musique	gold-nonfinal	0.663	0.007	0.739	0.018	0.001	0.003
musique	redundant	0.008	-0.005	0.020	0.007	0.001	0.003

Table 5: Form vs. content effect of a chain on p(gold), Qwen2.5-1.5B.

Dataset	Form (masked chain − no chain)	Content (real − masked chain)	n
clutrr	0.2 [-0.7,1.0]	0.8 [-3.6,4.8]	90
clutrr_copy	0.6 [-0.5,1.8]	75.1 [69.1,81.1]	90
musique	2.6 [-0.7,6.5]	56.4 [47.1,65.6]	80

8 Limitations

Assumptions. ?? 1?? 2 idealise the reader; we treat the propositions as testable predictions rather than guarantees, and the protocol is designed to falsify them. **Rollout-free interventions.** y_j^{tot} approximates do(H_j) by masking rather than re-generating descendants; we validate it against rollout PNS where compute allows. **Format dependence.** The DRG requires hop-structured chains linked by explicit mentions; pronouns break links and shrink $\mathcal{R}(y)$ to the terminal hop. We report format success, hop fidelity against the gold path and the fraction of fully connected DRGs, and fall back to CoT when parsing fails. **Closed-set scoring.** All CLUTRR methods, including CoT, are scored over the same 20 relations, so differences are not due to answer extraction. **Candidate sets.** Answer-level contrast needs a finite candidate set; it is exact on CLUTRR and approximate on MuSiQue (we report candidate recall). **Scale.** Main experiments use 1.5–7B models; larger models may rely less on shortcuts. **Data filter.** We use CLUTRR stories whose gold graph is a simple path; stories with back-edges contain known label ambiguities.

9 Conclusion

With interventions whose ground truth is known, we find that 1.5–1.7B models use their reasoning chains as a place to read an answer, not as a computation: the dependence appears exactly when

Table 6: Mediation of distractor and order effects on MuSiQue, Qwen2.5-1.5B (EM points).

Intervention	Method	TE	NDE (chain fixed)	NIE (via chain)
dist8	Direct (no CoT)	-3.3 [-10.0,2.5]	-4.2 [-10.0,0.9]	0.8 [-1.7,4.2]
dist8	CoT	-9.2 [-16.7,-2.5]	0.8 [-3.3,5.0]	-10.0 [-16.7,-4.2]
dist8	CAD	-0.8 [-7.5,5.8]	-1.7 [-7.5,4.2]	0.8 [-3.3,5.0]
dist8	AdaCAD	-0.8 [-8.3,5.8]	-2.5 [-9.2,3.3]	1.7 [-2.5,5.8]
dist8	CAD-chain	-5.8 [-13.3,0.8]	1.7 [0.0,4.2]	-7.5 [-15.0,-0.8]
dist8	Calibration (content-free)	-9.2 [-17.5,-1.7]	0.8 [0.0,2.5]	-10.0 [-18.3,-2.5]
dist8	Direct × CoT (PoE)	-8.3 [-15.8,-1.7]	1.7 [-3.3,6.7]	-10.0 [-17.5,-3.3]
dist8	HGCD (no graph)	-6.7 [-14.2,0.8]	2.5 [-0.8,5.8]	-9.2 [-16.7,-2.5]
dist8	HGCD (log-space)	-7.5 [-15.0,-0.8]	0.8 [0.0,2.5]	-8.3 [-15.8,-1.7]
dist8	HGCD (adaptive α)	-7.5 [-15.8,0.0]	1.7 [-1.7,5.0]	-9.2 [-16.7,-2.5]
dist8	**HGCD**	-7.5 [-15.0,0.0]	1.7 [-1.7,5.0]	-9.2 [-16.7,-2.5]
order	Direct (no CoT)	-4.2 [-10.8,2.5]	-3.3 [-9.2,2.5]	-0.8 [-4.2,2.5]
order	CoT	0.0 [-6.7,6.7]	-1.7 [-5.0,1.7]	1.7 [-5.0,8.3]
order	CAD	-4.2 [-11.7,3.3]	-1.7 [-8.3,5.0]	-2.5 [-7.5,2.5]
order	AdaCAD	-4.2 [-11.7,3.3]	-2.5 [-9.2,3.3]	-1.7 [-6.7,3.3]
order	CAD-chain	0.8 [-5.8,8.3]	0.0 [0.0,0.0]	0.8 [-5.8,8.3]
order	Calibration (content-free)	0.8 [-7.5,8.3]	0.0 [-2.5,2.5]	0.8 [-6.7,8.3]
order	Direct × CoT (PoE)	-3.3 [-10.0,4.2]	-3.3 [-9.2,3.3]	0.0 [-5.8,5.8]
order	HGCD (no graph)	4.2 [-3.3,11.7]	-0.8 [-2.5,0.0]	5.0 [-2.5,12.5]
order	HGCD (log-space)	0.0 [-7.5,6.7]	-0.8 [-2.5,0.0]	0.8 [-5.8,7.5]
order	HGCD (adaptive α)	2.5 [-5.8,10.8]	-2.5 [-5.8,0.0]	5.0 [-2.5,12.5]
order	**HGCD**	2.5 [-5.0,10.0]	-1.7 [-4.2,0.0]	4.2 [-3.3,11.7]

the answer is written in the chain and can be switched on by writing it there. Measuring this correctly requires graph-aware total-effect interventions, because single-step masking misses bridge reasoning entirely. Chain-of-thought moves a model's vulnerability to distractors from the readout to chain generation, which is why decoding rules that act on the readout, including the one we propose, do not help at this scale. Whether computation over the chain emerges with scale is the question our released protocol is designed to answer next.

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Table 7: CLUTRR accuracy (%), Qwen2.5-1.5B. ID: $k \leq 4$; OOD: $k \geq 5$.

Method	ID (n=118)	OOD (n=120)	All (n=238)
Direct (no CoT)	20.3 [13.6,28.0]	11.7 [5.8,17.5]	16.0 [11.3,20.6]
CoT	16.9 [11.0,23.7]	19.2 [12.5,26.7]	18.1 [13.9,23.5]
Self-consistency	22.9 [16.1,30.5]	19.2 [12.5,26.7]	21.0 [16.0,26.5]
CAD	20.3 [13.6,28.0]	17.5 [10.8,24.2]	18.9 [13.9,24.0]
AdaCAD	20.3 [13.6,27.1]	10.8 [5.0,16.7]	15.5 [10.9,20.2]
CAD-chain	18.6 [11.9,25.4]	17.5 [10.8,24.2]	18.1 [13.4,23.1]
Calibration (content-free)	21.2 [14.4,28.8]	26.7 [19.2,35.0]	23.9 [18.5,29.4]
Direct × CoT (PoE)	17.8 [11.0,24.6]	12.5 [6.7,19.2]	15.1 [10.9,19.7]
HGCD (naive mask)	20.3 [13.6,28.0]	25.8 [18.3,33.3]	23.1 [17.6,28.6]
HGCD (no graph)	16.9 [11.0,23.7]	19.2 [12.5,26.7]	18.1 [13.9,23.5]
HGCD (log-space)	17.8 [11.0,24.6]	27.5 [19.2,35.0]	22.7 [17.6,28.2]
HGCD (adaptive α)	16.9 [11.0,23.7]	19.2 [12.5,26.7]	18.1 [13.9,23.5]
HGCD (placebo)	18.6 [11.9,25.4]	21.7 [15.0,29.2]	20.2 [15.5,25.2]
HGCD	16.9 [11.0,23.7]	19.2 [12.5,26.7]	18.1 [13.9,23.5]
L1 + CoT	19.5 [12.7,26.3]	17.5 [10.8,25.0]	18.5 [13.9,23.5]
L1 + HGCD	19.5 [12.7,26.3]	17.5 [10.8,25.0]	18.5 [13.9,23.5]

Table 8: CLUTRR accuracy split by whether the model's chain reaches every gold-path entity.

Method	gold-path chain (n=41)	other chain (n=197)
CoT	17.1 [7.3,29.3]	18.3 [13.2,23.9]
Self-consistency	9.8 [2.4,19.5]	23.4 [17.3,29.4]
CAD-chain	19.5 [7.3,31.7]	17.8 [12.7,23.4]
HGCD (naive mask)	22.0 [9.8,34.1]	23.4 [17.3,28.9]
HGCD	17.1 [7.3,29.3]	18.3 [13.2,23.9]

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Table 9: MuSiQue exact match (%), Qwen2.5-1.5B.

Method	All (n=120)
Direct (no CoT)	25.0 [17.5,33.3]
CoT	20.8 [14.2,28.3]
CAD	24.2 [16.7,31.7]
AdaCAD	25.0 [17.5,33.3]
CAD-chain	21.7 [15.0,29.2]
Calibration (content-free)	25.0 [17.5,32.5]
Direct × CoT (PoE)	24.2 [16.7,32.5]
HGCD (naive mask)	24.2 [17.5,31.7]
HGCD (no graph)	21.7 [15.0,29.2]
HGCD (log-space)	23.3 [16.7,30.8]
HGCD (adaptive α)	21.7 [15.0,29.2]
HGCD (placebo)	21.7 [15.0,29.2]
HGCD	22.5 [15.8,30.0]
Free generation	23.3 [16.7,30.8]

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A Proofs

Proposition 1. y_j^{nai} changes only h_j and keeps $h_{j+1:k}$ at their observed values, so Δ_j^{nai} is by definition the controlled direct effect $\pi(a \mid h_j, h_{-j}) - \pi(a \mid \perp, h_{-j})$ [16]. Under assumption 1, $\pi(a \mid h_{1:k}) = g(a \mid h_{j+1:k})$ for $j < k$, so both terms equal $g(a \mid h_{j+1:k})$ and the difference is zero. y_j^{tot} additionally changes $h_{j+1:k}$ (their results are masked), so g is evaluated at a different argument and the difference is unconstrained. On compositional chains π reads the relation of every hop, so removing h_j alone changes the composed relation. □

Proposition 2. Substituting assumption 2 into eq. (1), with $\sum_j w_j = 1$ and π_s, π_d unaffected by interventions on hops: $s_\alpha(a) = (1{+}\alpha)[\lambda_c \pi_c(a \mid y) + \lambda_s \pi_s + \lambda_d \pi_d] - \alpha \sum_j w_j [\lambda_c \pi_c(a \mid y_j^{\text{tot}}) + \lambda_s \pi_s + \lambda_d \pi_d] = \pi(a \mid y) + \alpha \lambda_c \sum_j w_j [\pi_c(a \mid y) - \pi_c(a \mid y_j^{\text{tot}})]$. In log space, $\log(\lambda_c u + r) - \log(\lambda_c v + r)$ depends on $r = \lambda_s \pi_s + \lambda_d \pi_d$ unless $u = v$, so the shortcut terms do not cancel. □

Proposition 3. Let $m(a) = \pi(a \mid Q, P)$. CAD prefers a_d over g iff $(1{+}\alpha) \log \frac{m(a_d)}{m(g)} > \alpha \log \frac{\pi_s(a_d)}{\pi_s(g)}$. The left side is bounded below by $(1{+}\alpha) \log \frac{\lambda_d}{m(g)}$ since $m(a_d) \geq \lambda_d$, while the right side tends to $-\infty$ as $\pi_s(a_d) \to 0$. For HGCD, both terms of eq. (1) condition on the same P, so $\lambda_d \pi_d$ enters both and cancels (proposition 2). □

Proposition 4. Under exogeneity and monotonicity, Lemma 2 of Yu et al. [36] gives PNS = $P(A{=}a \mid \text{do}(S)) - P(A{=}a \mid \text{do}(S'))$. With $S = y$ and $S' = y_j^{\text{tot}}$ this is $\Delta_j^{\text{tot}}(a)$. Dividing eq. (1) by α and letting $\alpha \to \infty$ removes the $\pi(a \mid y)/\alpha$ term. □

B Implementation and Reproducibility

Candidate log-probabilities are computed with one prefix forward and a KV-cached pass over the candidate tokens (verified against full recomputation to 2×10^{-5}). Prompts are two-shot, hop-structured; the second CLUTRR shot uses a relation outside the label set to avoid leaking label statistics. All interventions are seeded functions of the example id. The Kaggle notebook embeds the full package and writes every table in this paper.

C SmolLM2-1.7B tables

Estimator	clutrr		clutrr_copy		musique	
	all	PS=1	all	PS=1	all	PS=1
Mask hop only (direct)	0.540	0.516	0.643	0.645	0.682	0.696
Mask hop + DRG descendants (total)	0.559	0.559	0.991	0.996	0.988	0.992
Replace result, keep downstream (direct)	0.503	0.474	0.651	0.643	0.658	0.658
Replace result, propagate (total)	0.532	0.494	0.996	1.000	0.987	0.993
DRG path indicator (structure only)	1.000	1.000	0.500	0.500	0.994	0.993
Placebo rewrite (control)	0.548	0.548	0.388	0.386	0.513	0.513
Mask token, content kept (control)	0.508	0.480	0.535	0.544	0.500	0.500

Dataset	Hop type	mask-tot	mask-dir	cf-tot	cf-dir	placebo	marker
clutrr	gold-final	0.010	0.010	0.063	0.063	0.050	0.006
clutrr	gold-nonfinal	0.019	0.008	0.027	0.016	-0.019	-0.017
clutrr	redundant	0.002	0.002	0.026	0.026	-0.003	-0.004
clutrr_copy	gold-final	0.632	0.632	0.686	0.686	-0.039	0.021
clutrr_copy	gold-nonfinal	0.685	0.011	0.706	-0.010	0.008	0.015
clutrr_copy	redundant	0.004	0.004	-0.010	-0.010	0.003	0.012
musique	gold-final	0.756	0.756	0.786	0.786	-0.026	-0.027
musique	gold-nonfinal	0.728	-0.036	0.746	0.005	0.005	0.027
musique	redundant	-0.028	-0.039	0.031	0.021	-0.008	0.018

Dataset	Form (masked chain − no chain)	Content (real − masked chain)	n
clutrr	-2.1 [-3.9,-0.4]	2.0 [-0.4,4.5]	90
clutrr_copy	-2.1 [-3.9,-0.2]	62.8 [57.5,67.9]	90
musique	-5.1 [-8.5,-2.1]	71.9 [65.7,77.9]	80

Intervention	Method	TE	NDE (chain fixed)	NIE (via chain)
dist8	Direct (no CoT)	-5.8 [-11.7,0.0]	-6.7 [-11.7,-1.7]	0.8 [-1.7,4.2]
dist8	CoT	-7.5 [-14.2,-0.8]	1.7 [-1.7,5.0]	-9.2 [-15.8,-2.5]
dist8	CAD	-4.2 [-9.2,0.8]	-5.0 [-9.2,-1.7]	0.8 [-1.7,4.2]
dist8	AdaCAD	-3.3 [-8.3,0.8]	-5.0 [-10.0,-0.8]	1.7 [-1.7,5.0]
dist8	CAD-chain	-7.5 [-14.2,-0.8]	1.7 [-1.7,5.0]	-9.2 [-15.8,-2.5]
dist8	Calibration (content-free)	-10.8 [-17.5,-3.3]	-1.7 [-5.0,1.7]	-9.2 [-15.8,-2.5]
dist8	Direct × CoT (PoE)	-7.5 [-14.2,0.0]	-0.8 [-4.2,1.7]	-6.7 [-13.3,0.0]
dist8	HGCD (no graph)	-8.3 [-15.0,-1.7]	0.8 [-2.5,4.2]	-9.2 [-15.8,-2.5]
dist8	HGCD (log-space)	-7.5 [-14.2,-0.8]	1.7 [-1.7,5.0]	-9.2 [-15.8,-2.5]
dist8	HGCD (adaptive α)	-7.5 [-14.2,-0.8]	1.7 [-1.7,5.0]	-9.2 [-15.8,-2.5]
dist8	**HGCD**	-8.3 [-15.0,-1.7]	0.8 [-2.5,4.2]	-9.2 [-15.8,-2.5]
order	Direct (no CoT)	-2.5 [-7.5,2.5]	-0.8 [-4.2,2.5]	-1.7 [-5.8,2.5]
order	CoT	2.5 [-5.0,10.0]	0.8 [-1.7,3.3]	1.7 [-5.8,10.0]
order	CAD	-0.8 [-5.0,3.3]	0.0 [-4.2,4.2]	-0.8 [-4.2,2.5]
order	AdaCAD	-0.8 [-4.2,2.5]	-0.8 [-4.2,2.5]	0.0 [-3.3,3.3]
order	CAD-chain	2.5 [-5.0,10.0]	0.8 [-1.7,3.3]	1.7 [-5.8,10.0]
order	Calibration (content-free)	0.0 [-7.5,7.5]	-0.8 [-2.5,0.0]	0.8 [-6.7,8.3]
order	Direct × CoT (PoE)	1.7 [-5.8,9.2]	-0.8 [-4.2,1.7]	2.5 [-4.2,10.0]
order	HGCD (no graph)	3.3 [-5.0,11.7]	0.8 [-1.7,3.3]	2.5 [-5.8,10.8]
order	HGCD (log-space)	2.5 [-5.0,10.0]	0.8 [-1.7,3.3]	1.7 [-5.8,10.0]
order	HGCD (adaptive α)	2.5 [-5.0,10.0]	0.8 [-1.7,3.3]	1.7 [-5.8,10.0]
order	**HGCD**	3.3 [-5.0,11.7]	0.8 [-1.7,3.3]	2.5 [-5.8,10.8]

Method	ID (n=118)	OOD (n=120)	All (n=238)
Direct (no CoT)	24.6 [16.9,32.2]	23.3 [15.8,30.8]	23.9 [18.5,29.8]
CoT	26.3 [18.6,34.7]	16.7 [10.8,23.3]	21.4 [16.4,26.9]
Self-consistency	33.1 [24.6,41.5]	20.0 [13.3,26.7]	26.5 [21.0,32.4]
CAD	24.6 [16.9,33.1]	22.5 [15.0,30.0]	23.5 [18.5,29.4]
AdaCAD	22.9 [16.1,30.5]	25.0 [17.5,32.5]	23.9 [18.9,29.4]
CAD-chain	26.3 [18.6,34.7]	16.7 [10.8,23.3]	21.4 [16.4,26.9]
Calibration (content-free)	30.5 [22.9,39.0]	19.2 [12.5,26.7]	24.8 [19.3,30.3]
Direct × CoT (PoE)	28.8 [20.3,37.3]	18.3 [11.7,25.0]	23.5 [18.1,29.0]
HGCD (naive mask)	24.6 [16.9,33.1]	17.5 [10.8,24.2]	21.0 [16.4,26.5]
HGCD (no graph)	24.6 [16.9,32.2]	18.3 [11.7,25.0]	21.4 [16.4,26.9]
HGCD (log-space)	26.3 [18.6,34.7]	16.7 [10.8,23.3]	21.4 [16.4,26.9]
HGCD (adaptive α)	26.3 [18.6,34.7]	16.7 [10.8,23.3]	21.4 [16.4,26.9]
HGCD (placebo)	26.3 [18.6,34.7]	16.7 [10.8,23.3]	21.4 [16.4,26.9]
HGCD	23.7 [16.1,31.4]	18.3 [11.7,25.0]	21.0 [16.0,26.5]
L1 + CoT	28.8 [20.3,37.3]	15.8 [10.0,22.5]	22.3 [17.2,28.2]
L1 + HGCD	24.6 [16.9,33.1]	18.3 [11.7,25.8]	21.4 [16.4,26.9]

D Pilot run (pipeline sanity check)

The following tables come from a CPU smoke run with Qwen2.5-0.5B-Instruct on a handful of examples. They verify that the pipeline runs end to end and are **not** evidence for or against any hypothesis.

Estimator	clutrr		musique	
	all	PS=1	all	PS=1
Δ^{naive} (mask hop only)	0.900	1.000	0.802	0.862
Δ^{total} (graph-aware mask)	0.881	1.000	1.000	1.000
DRG path indicator	1.000	1.000	1.000	1.000
Δ^{placebo} (content-preserving)	0.469	0.300	0.521	0.538
Δ^{marker} (mask token, content kept)	0.581	0.700	0.469	0.462

Dataset	Hop type	Δ^{total}	Δ^{naive}	Δ^{placebo}	Δ^{marker}
clutrr	gold-final	0.079	0.079	0.036	0.023
clutrr	gold-nonfinal	0.037	0.011	-0.048	-0.029
clutrr	redundant	-0.072	-0.072	-0.007	-0.018
musique	gold-final	0.682	0.682	-0.077	-0.062
musique	gold-nonfinal	0.423	-0.036	-0.050	-0.012
musique	redundant	-0.112	-0.112	-0.074	-0.018

Method	ID (n=8)	OOD (n=4)	All (n=12)
Direct (no CoT)	25.0 [0.0,62.5]	0.0 [0.0,0.0]	16.7 [0.0,41.7]
CoT	25.0 [0.0,62.5]	0.0 [0.0,0.0]	16.7 [0.0,41.7]
Self-consistency	12.5 [0.0,37.5]	0.0 [0.0,0.0]	8.3 [0.0,25.0]
CAD	25.0 [0.0,62.5]	0.0 [0.0,0.0]	16.7 [0.0,41.7]
AdaCAD	37.5 [12.5,75.0]	0.0 [0.0,0.0]	25.0 [0.0,50.0]
CAD-chain	25.0 [0.0,62.5]	0.0 [0.0,0.0]	16.7 [0.0,41.7]
HGCD (naive mask)	25.0 [0.0,62.5]	0.0 [0.0,0.0]	16.7 [0.0,41.7]
HGCD (no graph)	25.0 [0.0,62.5]	0.0 [0.0,0.0]	16.7 [0.0,41.7]
HGCD (log-space)	37.5 [12.5,75.0]	25.0 [0.0,75.0]	33.3 [8.3,58.3]
HGCD (adaptive α)	25.0 [0.0,62.5]	0.0 [0.0,0.0]	16.7 [0.0,41.7]
HGCD (placebo)	25.0 [0.0,62.5]	0.0 [0.0,0.0]	16.7 [0.0,41.7]
HGCD	25.0 [0.0,62.5]	0.0 [0.0,0.0]	16.7 [0.0,41.7]
L1 + CoT	25.0 [0.0,50.0]	0.0 [0.0,0.0]	16.7 [0.0,41.7]
L1 + HGCD	25.0 [0.0,50.0]	0.0 [0.0,0.0]	16.7 [0.0,41.7]

Even at this scale, the position breakdown shows the qualitative pattern predicted by proposition 1 on MuSiQue: naive masking leaves the answer probability of non-final gold hops nearly unchanged, whereas graph-aware masking reduces it substantially. With a few dozen hops this is an illustration, not a test.